

Integer Operation Rules

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Adding with the same signs: add and keep the sign. Adding with different signs: subtract and take the sign of the larger absolute value. Multiplying or dividing: same signs give a positive answer, different signs give a negative answer.

Round 1 — Retrieve It

1. When two numbers have the same sign, what do you do to add them?

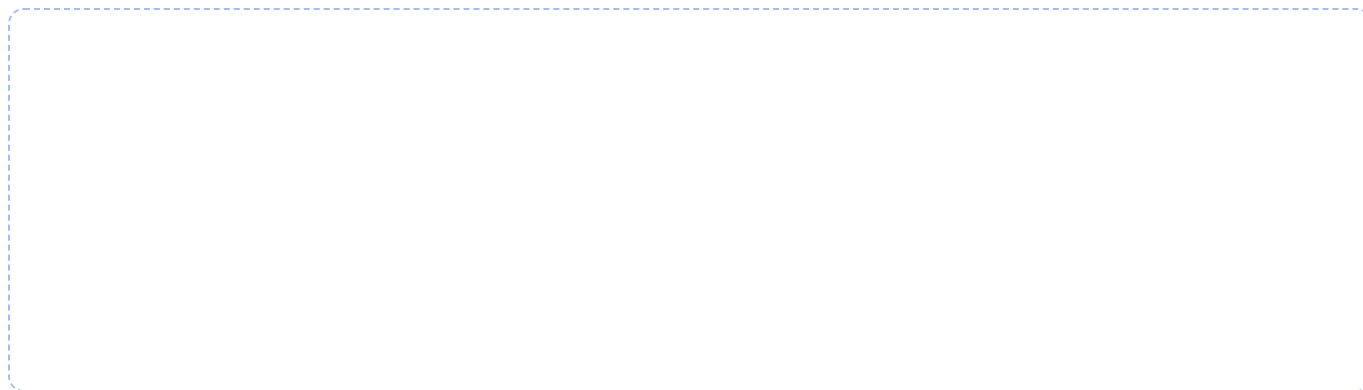
2. When two numbers have different signs, how do you decide the sign of the sum?

3. What is the sign of the answer when you multiply two numbers with the same sign?

4. What is the sign of the answer when you divide two numbers with different signs?

Round 2 — Sketch It

Draw a number line from -10 to 10 . Show $-3 + -5$ by starting at -3 and moving 5 units left. Circle where you land.



Integer Operation Rules — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Find $-7 + 4$.

2. Find $(-4) \times (-5)$.

3. Find $(-8) \div 2$.

Round 4 — Reason from Evidence

A student writes $-3 + -5 = 8$. Explain the mistake using the same-sign rule, then give the correct answer.

Integer Operation Rules — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how you decide the sign of a sum when the two numbers have different signs. Use one example that is not printed on the cue card.

C**Connect**

Connect the multiplication rule to the division rule. Why do $(-4) \times (-5)$ and its related division fact both follow the same-sign idea?

E**Extend**

Create your own problem that gives a negative product. Show the two factors and the finished multiplication.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Integer Operation Rules

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Add their absolute values and keep the shared sign.
2. Subtract the absolute values and take the sign of the number with the larger absolute value.
3. Positive.
4. Negative.

Sketch It — look for

A number line from -10 to 10 , starting at -3 and moving 5 units to the left, landing on and circling -8 ($-3 + -5 = -8$).

Apply It

1. $-7 + 4$: different signs, subtract $7 - 4 = 3$, take the sign of $-7 = -3$.
2. $(-4) \times (-5)$: same signs = positive = **20**.
3. $(-8) \div 2$: different signs = negative = **-4**.

Reason from Evidence — look for

Both addends are negative (same sign), so you add the absolute values and keep the negative sign. The correct answer is $-3 + -5 = -8$, not 8.

ACE — Extend look-for

Many answers possible. Check that the two factors have different signs so the product is negative and the multiplication is finished correctly.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.